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Assessment of seasonal variations in physico-chemical indicators of irrigation water affected by macroalgae-induced eutrophication

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To identify the factors responsible for eutrophication of the waters of two irrigation basins by the macroalgae *Cladophora glomerata*, present in a covered basin, and *Chara vulgaris*, present in an open basin, a novel periodic monitoring program was implemented in Morocco. A comparative analysis of the concentrations of different chemical elements in the water of the two basins was carried out within this framework. In order to account for the spatiotemporal variability of the measured parameters, the monitoring campaign, carried out from March 1, 2021, to March 1, 2022, relied on rigorous surveys and monthly sampling. The results indicated that nitrate concentrations in the first basin ranged from 3.08 ppm in winter to 4.72 ppm in summer, while in the second basin they ranged from 3.09 ppm in winter to 4.05 ppm in summer. Phosphorus concentrations in the first basin reached 6.78 ppm during winter and increased to 11.42 ppm during summer. By contrast, phosphorus levels were lower in the second basin, at 2.048 ppm in autumn and 3.5 ppm in summer. High concentrations of potassium were recorded in both tanks. In the *Cladophora glomerata* basin, potassium levels reached 146.16 ppm in winter and 191.66 ppm in summer. Concentrations were significantly higher in the *Chara vulgaris* basin, reaching 278.6 ppm in winter and 463.33 ppm in summer. The sustained presence of nutrients at high and relatively constant concentrations, combined with optimal sunlight conditions and poor maintenance practices, has encouraged the annual growth and persistence of these two species of macroalgae. Consequently, it is essential to implement pre-treatment measures before using this water for irrigation. It is also necessary to intervene at the source of these ponds and control the proliferation of harmful macroalgae to mitigate their negative impact on aquatic ecosystems.

Keywords Eutrophication, *Cladophora glomerata*, *Chara vulgaris*, Irrigation basin

Water resource management remains a critical challenge for Moroccan agriculture, as the sector consumes the largest share of the country's water supply¹. This challenge has intensified with the expansion of agricultural land to meet the rising demand for food, driven by population growth^{2, 3}. Nevertheless, the severe drought affecting the country has exacerbated the ongoing water crisis, posing a serious threat to agricultural production at both national and regional levels^{4, 5}. When rainfall is insufficient, irregular, and highly variable across regions, irrigation becomes essential to sustain crop growth⁶. Several irrigation methods and resources are available to agricultural producers^{2, 7}. In order to address these constraints, producers rely closed (covered) or open storage or irrigation basins, structured as underground, above-ground, or combined reservoirs^{8, 9}. These basins play a crucial role in water regulation by storing surpluses, limiting losses through evaporation and infiltration, and ensuring a reliable supply during critical crop growth periods⁶. Ensuring the water quality of these reservoirs is also essential for maintaining agricultural profitability and meeting national and international market standards¹⁰. However, the vulnerability of the Souss-Massa region to various sources of pollution negatively impacts the performance of irrigation basins^{11, 12}, despite their recognition as an effective solution for supplying

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high-quality irrigation water and sustaining agricultural productivity during drought periods¹³. Unfortunately, these water bodies are increasingly affected by eutrophication, characterised by varying levels of microalgal and macroalgal proliferation. The repercussions of this situation have had a detrimental effect on water quality in the reservoirs in question, causing blockages in the water pumping network¹⁴.

Indeed, this process is caused by the enrichment of the water body with nutrients, mainly nitrogen and phosphorus¹⁵. The most harmful aspects of this scourge are redundant, including the deoxygenation of water and the deterioration of its quality, which harms affected ecosystems (lack of transparency, odor development, and encroachment)¹⁶. Furthermore, this phenomenon affects all water surfaces, regardless of whether they are marine or freshwater, flowing or stagnant. The trophic status of these different water surfaces is determined by their nutrient content and biological production¹⁷. In general, the ageing process of water moves from an initial state of low nutrition (oligotrophic) to an intermediate state (mesotrophic) and finally to a final state of high nutrition (eutrophic)¹⁸. In addition, the stagnation of the waters is estimated to be a primary contributing factor to the acceleration of this process, which is evident in both natural and artificial water bodies, including ponds, lakes, dams, and irrigation basins¹⁹.

Furthermore, two macroalgae, *Cladophora glomerata* and *Chara vulgaris*, are key bioindicators of these phenomena in irrigation basins. The first species, *C. glomerata* is a filamentous green alga found in various types of water, whose growth is mainly caused by an increase in nutrients such as nitrogen and phosphorus in the water. *C. glomerata* belongs to the family Cladophoraceae, within the order Cladophorales of the phylum Chlorophyta. The taxonomic classification adopted in this study follows the systems established by Boedeker and Sviridenko²⁰, Hanyuda et al.²¹, and Soejima et al.²². The order Cladophorales is one of the few groups of green macroalgae capable of growing in both freshwater and marine environments. This is due to the structural and phenological traits possessed by its members, which enable adaptation to these two habitats²³. Its cells are cylindrical, multinucleate and have cell walls, with a layer of cuticle on the outside that gives them a rigidity and mechanical strength²³. The second species, *C. vulgaris*, is classified within the family Characeae, order Charales, phylum Charophyta. This taxonomic placement is based on classifications reported in the literature^{24, 25, 26, 27, 28}. Members of the Characeae are able to colonise a wide range of freshwater and brackish habitats, whether permanent or temporary. These are regarded as the most evolutionarily advanced green algae and are considered to be the closest relatives of terrestrial plants (embryophytes)²⁹. Characeae are characterised by their complex structure, with branched green stems, nodes, internodes and specialised reproductive structures³⁰. Some species are particularly sensitive, especially to increases in nutrient levels, which is why they are excellent ecological indicators (sentinel organisms) of the functioning of aquatic environments and changes in their quality^{31, 32}.

In fact, in the Souss-Massa region, characterised by low rainfall and limited access to non-conventional water resources, the authorities have implemented strategies to optimise water use³³. The region's irrigation basins are supplied by two primary sources: the Youssef Ben Tachafine Dam and groundwater. While these resources do ensure a relatively stable flow, they remain highly vulnerable to pollution, which poses significant challenges to the sustainable management of irrigation water^{12, 34}. This situation highlights how eutrophication, in combination with irrigation practices and water source quality, directly affects the ecological conditions of the basins and promotes the growth of macroalgae.

In light of the established value of these two species, their successful integration across multiple sectors, and the findings from the initial phase of this research¹⁴, our focus was specifically on them, with the objective of subsequently utilising them in agriculture as biofertilisers and biostimulants. Given that each species inhabits a separate pond, the primary objective of the present study was to identify the physicochemical factors underlying their emergence and persistence. This original comparative study of the two environments allowed us to identify the key factors likely responsible for the relatively stable abundance of these two macroalgae. Regular monitoring was conducted in the first covered basin, where *Cladophora glomerata* was collected, and in the second open basin, where *Chara vulgaris* was sampled. This systematic study, which was conducted from 1 March 2021 to 1 March 2022, revealed that increasing climate variability, insufficient reservoir maintenance, and high nutrient inputs – particularly nitrates, phosphorus, potassium, magnesium, and manganese – from historically contaminated sources have played a key role in the long-term persistence of macroalgal blooms. Further opportunities may be explored based on these findings, including the large-scale cultivation of these two species for various future applications in upcoming projects.

Materials and methods

Study area and sampling

The irrigation basins examined in this study are situated in the Souss Massa region. The first, a covered basin known as Douar Ghzala, is dominated by *C. glomerata*, while the second, an open basin called Salad Time, is characterised by *C. vulgaris* (Fig. 1).

As part of the annual hydrological study, samples were collected at monthly intervals from 1 March 2021 to 1 March 2022. The samples were meticulously scheduled to encompass the entire annual cycle, with samples collected in the first week and last week of each month. A methodology was implemented in order to capture the variations between the surface and the bottom of each basin. The methodology entailed the collection of three surface samples and three deep samples at the inlet and outlet of the basin. At the study site, attention was first focused on determining the intensity of the colour and odour of the water, which are the first two organoleptic criteria.

Physicochemical analysis

The hydrogen potential (pH) was measured using a pH meter (HI-8014) equipped with a specific ORICON electrode. The water temperature (°C) was recorded using a multiparameter thermometer (W.T.W. LF318), and the conductivity (mS/cm) was assessed using a conductivity meter (W.T.W. LF 318, Tetracon 325 measuring



Fig. 1. (a) Open pond “Salad Time” showing eutrophication by the macroalga *Chara vulgaris*; (b) Covered pond “Douar Ghzala” characterised by dense mats of the macroalga *Cladophora glomerata*. Photo credit: Author.

cell). As part of the chemical analysis of the water samples, these were meticulously stored in 500 ml opaque glass bottles. These containers, meticulously maintained at a low temperature (4 °C), were expeditiously conveyed to three laboratories: two situated in Agadir (namely, the INRA [National Institute for Agricultural Research] water and soil analysis laboratory and the IVA [Hassan II Agronomic and Veterinary Institute] water, soil and plant analysis laboratory) and one in Rabat at the National Laboratory for Pollution Studies and Monitoring (LNESP, Rabat), specifically the micropollutant analysis department, under the oversight of the Ministry of Energy Transition and Sustainable Development. All analytical equipment used in this study was carefully calibrated in accordance with the procedures outlined in the Guide (2009). Thereafter, physicochemical and mineral analyses were performed in strict compliance with the standardised methods described by Rodier et al.³⁵, with the aim of accurately characterising the mineral composition of the water in both basins.

In fact, A 50 ml water sample was dried at 105 °C to determine total dissolved solids (TDS) (NF T90-029:1970), and then calcined to assess organic matter content (LOI) (NF EN 15935). Carbonates and bicarbonates were titrated using phenolphthalein and helianthine, respectively (ISO 9963-1:1996). Phosphorus levels (660 nm) were measured using continuous flow analysis (CFA) with ascorbic acid (NM ISO 15681-2:2007), while nitrate levels (540 nm) were determined via reduction to nitrites using an active cadmium column and a reagent mixture containing sulfanilamide and α -naphthylamine dihydrochloride (NM ISO 13395:1999). Chlorides were analyzed via argentometry (Mohr's method) (NM ISO 9297:2014), and sulfate concentrations were measured turbidimetrically by precipitating sulfate ions as barium sulfate (BaSO_4) in a hydrochloric acid medium, with turbidity readings taken at 420 nm (ISO 3431:1976). Potassium and sodium levels were assessed using a flame spectrophotometer, measuring Na at 589 nm and K at 769.9 nm (ISO 9964-2:1993). Calcium and magnesium concentrations were determined via atomic absorption spectrometry at 422.67 nm and 202.58 nm, respectively (ISO 7980:1986). Boron content was assessed at 420 nm using the azomethine-H method (ISO 9390:1990). Knowing that all parameters required optical density (OD) readings, ranges were prepared in proportion to the type of dosage^{36, 37, 38, 39, 40}.

Trace metal elements (TMEs) analysis

Physicochemical analyses and trace metal determinations were performed following standardized protocols to ensure the reliability and reproducibility of the results. The studied metals were considered as trace metal elements (TMEs), including essential elements (Fe, Mn, Zn, Cu) and potentially toxic elements (Cd, Pb, Ni, Al), rather than strictly classified “heavy metals”. The concentrations of iron (Fe) (248.33 nm), manganese (Mn) (279.48 nm), and copper (Cu) (324.78 nm) were determined using flame atomic absorption spectrometry (FAAS), in accordance with standard procedures described in the Standard Methods for the Examination of Water and Wastewater⁴¹. Trace metal element concentrations were also measured following international standards, including (ICP-OES 2007), ISO 17294-2:2016 (ICP-MS 2016), and ISO 15586:2003 (AAS 2003). EPA Methods 200.7 (1994) and 200.8 (1994)^{42, 43, 44} served as additional references for trace element analysis in aqueous matrices. All procedures were performed under controlled conditions to minimize contamination and maintain analytical precision. For the determination of the element content of metallic traces (TME) Cadmium (Cd, 228.802 nm), zinc (Zn, 472.215 nm), nickel (Ni, 341.476 nm), lead (Pb, 405.781 nm), and aluminium (Al, 396.152 nm) a test portion of a water sample, which can contain up to 20 g/l of suspended solids and up to 5 g/l of total organic carbon (TOC), is mixed with nitric acid 65% and heated in a microwave oven to a temperature of 170 °C, for a well-defined period. During the analysis and for the preparation of reagents that must be of good quality or equivalent, we use only ultra-pure or equivalent water (Resistivity equal to 18.3 M Ω) (NM ISO 15587-2 and EPA 3015A⁴⁰). This analysis was carried out by Microwave Atomic Emission Spectrometry using the device the Agilent 4210 MP-AES, at the National Laboratory of Studies and Pollution Monitoring (LNESP, Rabat), of service of Micropollutant Analyses, under the direction of the Ministry of Energy Transition and Sustainable Development (Element 2007; ISO 2002).

Macroalgae strains identification

To assess the condition of the two algal species, samples were collected from each basin and preserved separately in opaque glass bottles using a 0.5% Lugol's iodine solution. In addition, Lugol's iodine was used for temporary microscopic observation of macroalgal tissues, enhancing contrast and allowing the visualization of cellular structures such as chloroplasts and reproductive elements. However, it was not used as a long-term preservation method^{45, 46}. The bottles were then placed in a refrigerator in the molecular biology laboratory at the National Institute for Agronomic Research (INRA), set to 4 °C. Microscopic observations were performed using an Olympus BX53M optical microscope with Toopview software at 400× magnification (40× objective lens×10× eyepiece) in the molecular biology laboratory of the National Institute of Agronomic Research (INRA)¹⁴. Algal identification followed the guidelines of Michalak and Messyas²³, Laplace-Tretyure et al.²⁷, Auderset Joye and Schwarzer³⁰, Lauret et al.⁴⁷, with additional consultation of international databases⁴⁸.

In addition, Fig. 2 provides a comprehensive illustration of the morphological and microscopic characteristics that distinguish the two species, thereby demonstrating that each possesses traits that are distinctly specific to its genus.

The texture of *C. glomerata* is rough, with long, light green, unbranched filaments 0.3 to 1 mm in diameter. The sections dividing the filaments can be seen with the naked eye. The algae can be covered with diatoms and are dark green or yellow. The texture of *C. vulgaris* is somewhat rough, consisting of a fixed axis in the sediment interspersed with whorls of twigs placed at regular intervals. The whorls are closer to the upper end of the stem where the twigs are clumped. Each of its branches acting as a leaf is called phylloid where specialised reproductive structures are noticed (oogonia and antheridia).

Statistical analysis

A one-way analysis of variance (ANOVA) was conducted to assess the impact of seasonal variation on chemical concentrations in each basin. This approach enabled comparison of mean values across multiple independent seasonal groups while accounting for within-group variability. When significant differences were observed, Duncan's multiple range post hoc test was applied at a 5% significance level to determine pairwise differences between seasonal means. The F-statistic, representing the ratio of between-group (inter-seasonal) variance to within-group (intra-seasonal) variance, was used to evaluate the significance of seasonal effects on the measured variables. Additionally, Principal Component Analysis (PCA) was performed to investigate relationships among physico-chemical and metallic variables and to characterize the overall structure of the dataset. Principal Component Analysis (PCA) was conducted using a centered and scaled correlation matrix. The data matrix included six individuals, representing monthly water samples from the two basins, and 24 active variables, which comprised 19 physicochemical parameters and five trace metal concentrations. Standardization by centering and scaling was systematically applied to mitigate scale effects arising from differing units of measurement (e.g., ppm for metals vs. µS/cm for conductivity). This preprocessing step ensured that each environmental parameter contributed equally to the total inertia and to the definition of the first and second principal component axes. This multivariate technique facilitated data dimensionality reduction and summarized the principal patterns of variation among samples^{49, 50}. All statistical processing was performed using IBM SPSS Statistics, version 25.

Results

The results obtained from the annual monitoring of the two basins revealed marked seasonal fluctuations in water quality. The first basin (Table 1) corresponds to the Douar Ghzala site, where the dominant sampled alga was *C. glomerata*, whereas the second basin (Table 2) corresponds to the Salade Time pond, characterised by the presence of *C. vulgaris*. The physicochemical and metallic analyses conducted in these two environments point to pronounced temporal dynamics across the different parameters measured. Prior to multivariate analysis, data normality was assessed using the Kolmogorov–Smirnov test, which showed no statistically significant deviation from normality ($p > 0.05$). This validation step justified the subsequent use of parametric statistical procedures,

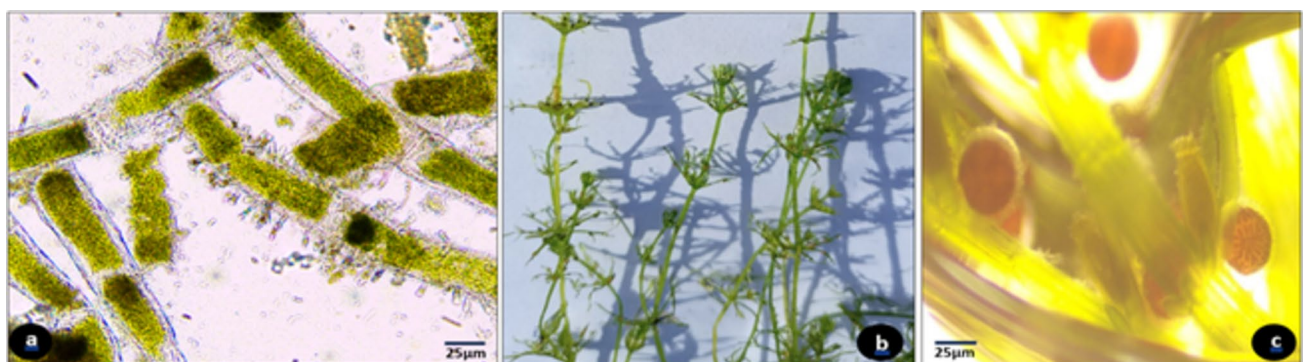


Fig. 2. (a) *Cladophora glomerata* filaments surrounded by *Gomphonema micropus*. Live specimen, wet mount, 400x magnification, scale bar: 25 micrometres, (b) whorled twigs *Chara vulgaris* (length >15cm), (c) oogonia and antheridia of *Chara vulgaris*. Live specimen, wet mount, 400x magnification, scale bar: 25 micrometres. Photo credit: Author.

Parameters	Units	Analysis periods					
		Spring period (03/04/05)	Summer period (06/07/08)	Autumn period (09/10/11)	Winter period (12/01/02)	F (3,8)	p-value
Color	-	Green- Dark green	Dark green	Dark green	Green	-	-
Smell	-	Nauseating musty odour	Septic tank	Septic tank	Pungent odour	-	-
pH	-	7.87 ± 0.00577 ^a	8.03 ± 0.00577 ^d	7.97 ± 0.00577 ^c	7.92 ± 0.00577 ^b	2145.31	<0.001
Temperature	(°C)	22 ± 0.00577 ^a	23.11 ± 0.00577 ^c	23.01 ± 0.00577 ^b	23.01 ± 0.00577 ^b	18241.66	<0.001
Conductivity	(mS/cm)	2.86 ± 0.00493 ^b	2.98 ± 0.00577 ^c	2.83 ± 0.00577 ^a	3.02 ± 0.00577 ^d	35698.44	<0.001
Total dry residue	g/l	0.67 ± 0.00318 ^a	0.745 ± 0.00088 ^d	0.713 ± 0.00120 ^c	0.685 ± 0.00058 ^b	12754.29	<0.001
Loss on ignition (LOI)	%	19.67 ± 0.00577 ^b	21.13 ± 0.00577 ^d	20.895 ± 0.00058 ^c	19.47 ± 0.00577 ^a	9643.57	<0.001
Carbonates	ppm	70 ± 0.00882 ^b	78 ± 0.00577 ^d	74 ± 0.00577 ^c	69 ± 0.00577 ^a	14582.11	<0.001
Bicarbonates		2401.63 ± 0.0057 ^b	2806.73 ± 0.00577 ^d	2786.62 ± 0.00577 ^c	2306.53 ± 0.00577 ^a	48902.35	<0.001
Phosphorus P2O5		8.32 ± 0.0057 ^b	11.42 ± 0.0057 ^d	9.3 ± 0.0057 ^c	6.78 ± 0.0057 ^a	22531.88	<0.001
Nitrates NO ₃		3.4 ± 0.0057 ^b	4.72 ± 0.0057 ^d	4.1 ± 0.0057 ^c	3.08 ± 0.00441 ^a	31884.77	<0.001
Chlorides Cl		150.88 ± 0.0057 ^b	170.16 ± 0.0057 ^d	160.85 ± 0.0057 ^c	140.8 ± 0.0057 ^a	45221.19	<0.001
Sulphate SO ₄		256.09 ± 0.0057 ^b	340 ± 0.0057 ^d	278.7 ± 0.0057 ^c	216.7 ± 0.0057 ^a	51104.66	<0.001
Potassium K		177.76 ± 0.0057 ^b	191.66 ± 0.0057 ^d	180.96 ± 0.0057 ^c	146.16 ± 0.0057 ^a	27884.55	<0.001
Sodium Na		52.23 ± 0.0057 ^b	68.46 ± 0.0057 ^d	58.34 ± 0.0057 ^c	48.65 ± 0.0057 ^a	30144.92	<0.001
Calcium Ca		122.17 ± 0.0057 ^b	130 ± 0.0057 ^d	125.7 ± 0.0057 ^c	120.3 ± 0.0057 ^a	26544.80	<0.001
Magnesium Mg		220.2 ± 0.0057 ^b	296.9 ± 0.0057 ^d	251.9 ± 0.0057 ^c	178.5 ± 0.0057 ^a	33218.64	<0.001
Manganese Mn		15.78 ± 0.0057 ^b	27.183 ± 0.00058 ^d	19.45 ± 0.0057 ^c	10.3 ± 0.0057 ^a	21754.18	<0.001
Iron Fe		0.123 ± 0.00058 ^a	0.33 ± 0.0057 ^c	0.268 ± 0.00088 ^b	0.11 ± 0.0057 ^a	19551.37	<0.001
Copper Cu		0.00 ± 0.00 ^a	0.00 ± 0.00 ^a	0.002 ± 0.00006 ^b	0.00 ± 0.00 ^a	555.56	0.08
Boron B		0.0012 ± 0.00006 ^b	0.00 ± 0.00 ^a	0.002 ± 0.00006 ^c	0.00 ± 0.00 ^a	533.33	<0.001
Zinc Zn		0.118 ± 0.00058 ^b	0.351 ± 0.00058 ^d	0.221 ± 0.00058 ^c	0.101 ± 0.00058 ^a	17444.28	<0.001
Cadmium Cd		0.0018 ± 0.00006 ^a	0.0033 ± 0.00009 ^c	0.0028 ± 0.00006 ^b	0.0037 ± 0.00006 ^d	15411.02	<0.001
Nickel Ni		0.005 ± 0.00058 ^b	0.014 ± 0.00058 ^c	0.012 ± 0.00058 ^c	0.002 ± 0.00058 ^a	13874.55	<0.001
Lead Pb		0.00 ± 0.00 ^a	0.0032 ± 0.00006 ^c	0.0011 ± 0.00006 ^b	0.00 ± 0.00 ^a	12155.93	<0.001
Aluminium Al		1.014 ± 0.00058 ^a	4.213 ± 0.00058 ^d	3.51 ± 0.00882 ^c	1.21 ± 0.00577 ^b	40611.27	<0.001

Table 1. Variability of the physico-chemical parameters of the water in the covered irrigation basin of *Cladophora glomerata* “Douar Ghzala” according to the seasons. Mean value ± standard error (mS/cm: millisiemens per centimeter; ppm: part per million). F: Fisher’s F test. Means in the same row followed by different superscript letters (a–d) differ significantly at $p < 0.05$ according to Duncan’s Multiple Range Test. Letters are assigned in ascending order of concentration, such that $d > c > b > a$.

including principal component analysis. The PCA results reveal a particularly clear seasonal structuring of the two basins (Fig. 3: (a) *C. glomerata* basin; (b) *C. vulgaris* basin). Seasonal projections show a distinct separation between warm and cold periods, thereby confirming the major influence of climatic conditions on the physicochemical composition of the water. In both basins, the highest values for most of the studied parameters were recorded during summer, followed by autumn, indicating a pronounced seasonal intensification of the chemical processes affecting water quality.

The hydrogen potential (pH) of the *C. glomerata* basin ranged between a minimum of 7.87 in the spring phase and a maximum of 8.03 in the summer phase. Its temperature varied between 22 °C in spring and 23.11 °C in summer, while conductivity fluctuated between 2.83 mS/cm in autumn and 3.02 mS/cm in winter. In the *C. vulgaris* basin, higher values were recorded in summer: pH 8.07, conductivity 6.4 mS/cm, and a temperature of 23.4 °C. The lowest pH was 7.75 in autumn, with a minimum temperature of 22.1 °C in spring and conductivity at 3.23 mS/cm in autumn. Regarding the physical water properties of the two basins, the color of the *C. glomerata* basin was generally dark green from June to November and emitted a foul smell. During winter, the water changed from green with a pungent smell to dark green with a nauseating musty odor at the end of spring. In the *C. vulgaris* basin, the water appeared slightly dark green with a sulfurous smell during the summer and autumn periods, changing to a greener color with a milder sulfur odor for the rest of the year. Our findings indicate that the concentration of dry residues was lowest in spring 0.67 g/L and highest in summer 0.74 g/L in the *C. glomerata* basin. In the *C. vulgaris* basin, these concentrations were higher, reaching 0.70 g/L in winter and 1.5 g/L in summer. Furthermore, organic matter levels in the *C. vulgaris* basin were higher in summer 22.52% than in the *C. glomerata* basin during the same season 21.13%. The lowest values were recorded in winter, reaching 19.56% in the open *C. vulgaris* basin and 19.47% in the covered *C. glomerata* basin. The analysis of carbonates and bicarbonates demonstrated that, within the *C. glomerata* basin, the lowest concentrations were recorded during the winter months (69 ppm and 2,306.53 ppm, respectively), while the highest levels were observed during the summer months (78 ppm and 2,806.73 ppm, respectively). The *C. vulgaris* basin exhibited even higher values, both in winter (78.56 ppm and 2,807.05 ppm) and in summer (114.02 ppm for carbonates and 2,904.36 ppm for bicarbonates), indicating a marked increase compared with the seasonal values recorded

Parameters	Units	Analysis periods					
		Spring period (03/04/05)	Summer period (06/07/08)	Autumn period (09/10/11)	Winter period (12/01/02)	F (3,8)	p-value
Color	-	Green	Slightly dark green	Slightly dark green	Green	-	-
Smell	-	Slightly sulphurous	Sulphurous odour	Sulphurous odour	Slightly sulphurous	-	-
pH	-	7.78 ± 0.00351 ^b	8.07 ± 0.00521 ^d	7.75 ± 0.00577 ^a	7.96 ± 0.00577 ^c	872.37	<0.001
Temperature	(°C)	22.1 ± 0.00577 ^a	23.4 ± 0.00577 ^c	23.1 ± 0.00577 ^b	23.1 ± 0.00088 ^b	12,808.19	<0.001
Conductivity	(mS/cm)	4.40 ± 0.00577 ^b	6.4 ± 0.00577 ^d	3.23 ± 0.00577 ^a	5.47 ± 0.00577 ^c	55,994.75	<0.001
Total dry residue	g/l	0.76 ± 0.00577 ^b	1.5 ± 0.00577 ^d	0.863 ± 0.00088 ^c	0.709 ± 0.00058 ^a	10,444.51	<0.001
Loss on ignition (LOI)	%	20.42 ± 0.00577 ^b	22.522 ± 0.00058 ^d	21.6 ± 0.00577 ^c	19.56 ± 0.00577 ^a	8233.16	<0.001
Carbonates	ppm	109 ± 0.00577 ^c	114.02 ± 0.00577 ^d	102.4 ± 0.00577 ^b	78.56 ± 0.00577 ^a	13,244.20	<0.001
Bicarbonates		2898.15 ± 0.00577 ^b	2904.36 ± 0.00577 ^d	2898.25 ± 0.00577 ^c	2807.05 ± 0.00577 ^a	46,601.52	<0.001
Phosphorus P2O5		2.89 ± 0.00088 ^c	3.51 ± 0.00577 ^d	2.048 ± 0.00033 ^a	2.13 ± 0.00088 ^b	19,851.77	<0.001
Nitrates NO ₃		3.11 ± 0.00577 ^b	4.05 ± 0.00115 ^d	3.89 ± 0.00173 ^c	3.09 ± 0.00145 ^a	28,774.22	<0.001
Chlorides Cl		397.04 ± 0.00273 ^b	425.4 ± 0.00120 ^d	406.49 ± 0.00153 ^c	323.8 ± 0.00088 ^a	42,105.18	<0.001
Sulphate SO ₄		256.1 ± 0.00577 ^b	387.5 ± 0.00058 ^d	301 ± 0.00265 ^c	229 ± 0.00577 ^a	49,661.09	<0.001
Potassium K		321 ± 0.00577 ^b	463.33 ± 0.00577 ^d	341.34 ± 0.00577 ^c	278.6 ± 0.00577 ^a	26,174.08	<0.001
Sodium Na		21.3 ± 0.00577 ^b	27.38 ± 0.00577 ^d	25.4 ± 0.00577 ^c	17.8 ± 0.00577 ^a	28,555.41	<0.001
Calcium Ca		112.4 ± 0.00577 ^b	139.16 ± 0.00577 ^d	132.5 ± 0.00577 ^c	109.5 ± 0.00577 ^a	24,114.64	<0.001
Magnesium Mg		312.3 ± 0.00577 ^b	365.95 ± 0.00577 ^d	333.5 ± 0.00577 ^c	298.7 ± 0.00577 ^a	31,105.27	<0.001
Manganese Mn		7.56 ± 0.00577 ^b	10.34 ± 0.00577 ^d	9.770 ± 0.00577 ^c	5.06 ± 0.00577 ^a	20,155.88	<0.001
Iron Fe		0.23 ± 0.00088 ^b	0.582 ± 0.00058 ^d	0.34 ± 0.00115 ^c	0.19 ± 0.00120 ^a	17,401.95	<0.001
Copper Cu		0.00 ± 0.00 ^a	0.00 ± 0.00 ^a	0.00 ± 0.00 ^a	0.00 ± 0.00 ^a	0.00	>0.05
Boron B		0.002 ± 0.00012 ^b	0.035 ± 0.00027 ^d	0.015 ± 0.00028 ^c	0.0009 ± 0.00006 ^a	139,853.24	<0.001
Zinc Zn		0.023 ± 0.00006 ^a	0.221 ± 0.00067 ^d	0.123 ± 0.00058 ^c	0.032 ± 0.00058 ^b	17,444.28	<0.001
Cadmium Cd		0.0009 ± 0.00006 ^b	0.002 ± 0.00006 ^c	0.0009 ± 0.00006 ^b	0.00 ± 0.00 ^a	15,411.02	<0.001
Nickel Ni		0.0039 ± 0.00006 ^c	0.0038 ± 0.00006 ^c	0.0019 ± 0.00003 ^b	0.00 ± 0.00 ^a	13,874.55	<0.001
Lead Pb		0.003 ± 0.00006 ^b	0.009 ± 0.00006 ^d	0.004 ± 0.00006 ^c	0.00 ± 0.00 ^a	12,155.93	<0.001
Aluminium Al		0.247 ± 0.00058 ^b	0.447 ± 0.00058 ^d	0.337 ± 0.00058 ^c	0.127 ± 0.00058 ^a	40,611.27	<0.001

Table 2. Variability of the physicochemical parameters of the water in the open irrigation basin of Chara vulgaris “Salad Time” according to the seasons. Mean value ± standard error. (mS/cm: millisiemens per centimeter; ppm: part per million). F: Fisher’s F test. Means in the same row followed by different superscript letters (a–d) differ significantly at $p < 0.05$ according to Duncan’s Multiple Range Test. Letters are assigned in ascending order of concentration, such that $d > c > b > a$.

in the *C.glomerata* basin. Furthermore, the report highlights the phosphorus content of the *C.vulgaris* basin, reaching 3.51 ppm in summer and 2.89 ppm in spring, before declining to 2.04 ppm in autumn. In contrast, the *C.glomerata* basin exhibited the highest values throughout the study period, with 8.32 ppm in spring, 11.42 ppm in summer, followed by a decrease to 6.78 ppm in winter. Nitrate concentrations remained relatively stable in both basins during the winter season, with values of 3.08 ppm in the *C.glomerata* basin and 3.09 ppm in the *C. vulgaris* basin. However, during the summer months, these values increased, reaching 4.72 ppm and 4.05 ppm respectively in the two basins. Chloride analysis revealed that the *C.vulgaris* pond had significantly higher concentrations than those recorded for the second pond. Indeed, these values reached 323.8 ppm in winter and 425.4 ppm in summer. In contrast, the *C.glomerata* pond exhibited significantly lower levels, with values reaching only 140.8 ppm in winter and 170.16 ppm in summer. Analyses revealed high concentrations of sulphates, particularly in the *C.vulgaris* basin, with values ranging from 216.7 ppm to 340 ppm in the *C. glomerata* basin and from 229 ppm to 387.5 ppm in the *C.vulgaris* basin, indicating a seasonal increase from winter to summer. A notable increase in potassium concentrations was observed in the *C.vulgaris* pond, ranging from 278.6 ppm in winter to 463.33 ppm in summer. In contrast, the *C.glomerata* pond exhibited lower potassium levels, ranging from 146.16 ppm in winter to 191.66 ppm in summer. Conversely, sodium concentrations in the *C.glomerata* pond were more than double those in *C.vulgaris*, with 48.65 ppm in winter and 68.46 ppm in summer, compared to 17.8 ppm and 27.38 ppm, respectively, in the *C.vulgaris* pond. With regard to calcium levels, the *C.glomerata* reservoir registered 120.3 ppm during the winter months and 130 ppm in the summer period. In comparison, the *C.vulgaris* reservoir recorded 109.5 ppm and 139.16 ppm respectively over the same periods. The highest magnesium levels were recorded in the open *C.vulgaris* pond, with 298.7 ppm in winter and 365.95 ppm in summer. The *C.glomerata* pond had lower concentrations, with 178.5 ppm in winter and 296.9 ppm in summer. Metalloid boron (B) concentrations exhibited significant spatial and seasonal variability across the studied basins. The highest levels were recorded during the summer, specifically in the *C.vulgaris* pond, which reached 0.035 ppm. In contrast, the *C.glomerata* pond showed substantially lower concentrations, with 0.002 ppm in autumn and falling below the detection limit (0.000 ppm) during the winter. In the latter season, the *C.vulgaris* pond maintained a trace boron level of 0.0009 ppm.

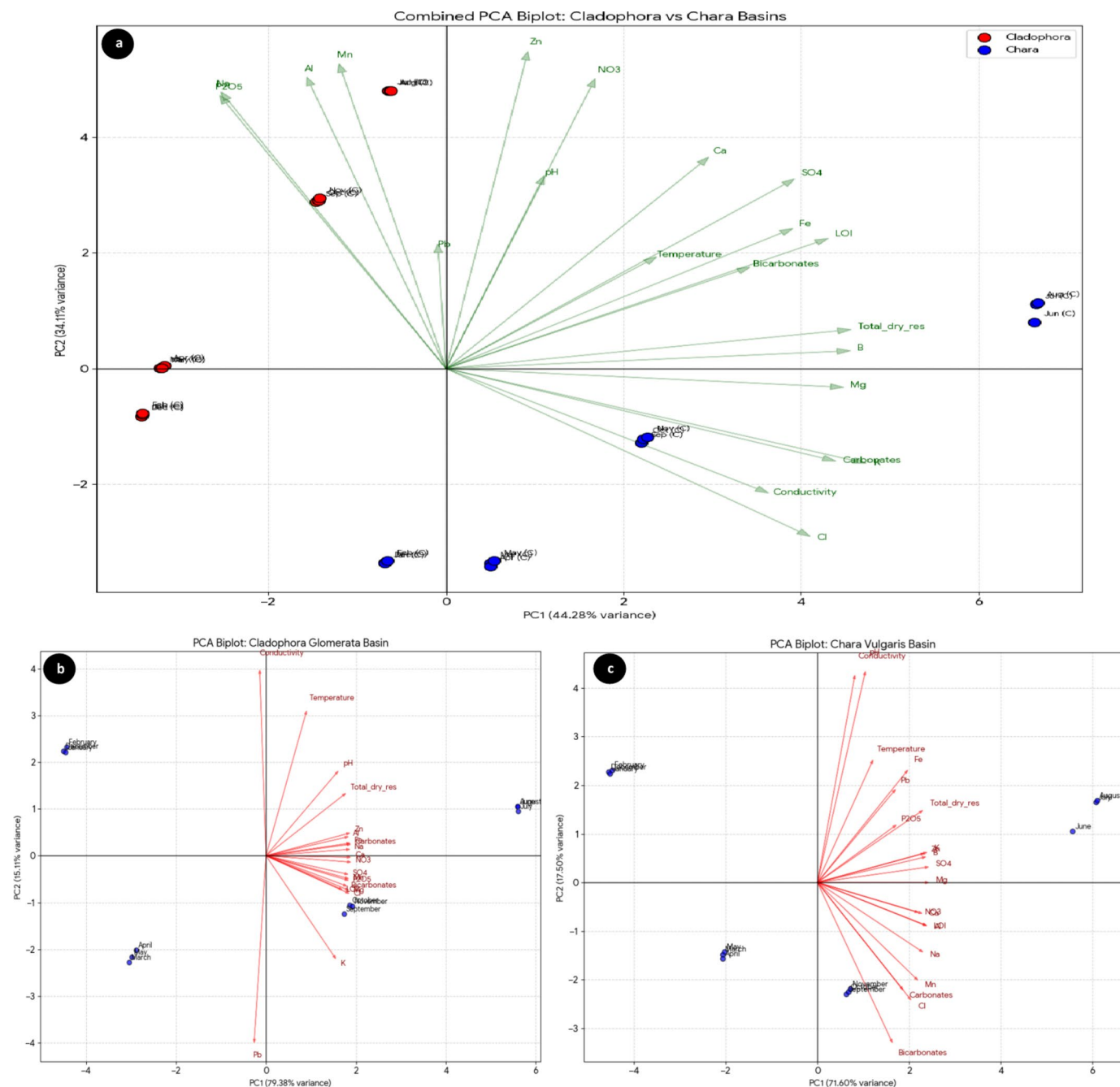


Fig. 3. Principal component analysis (PCA) of physicochemical variables. (a) Comparison between *Cladophora glomerata* basin and *Chara vulgaris* basin; (b) PCA of physicochemical variables of the *Cladophora glomerata* basin; (c) PCA of physicochemical variables of the *Chara vulgaris* basin. Variables included: temperature, electrical conductivity, total dry residue, loss on ignition (LOI), carbonates, bicarbonates, phosphorus (P_2O_5), nitrates (NO_3^-), chlorides (Cl^-), sulphates (SO_4^{2-}), potassium (K), sodium (Na), calcium (Ca), magnesium (Mg), manganese (Mn), iron (Fe), copper (Cu), boron (B), zinc (Zn), cadmium (Cd), nickel (Ni), lead (Pb), and aluminium (Al).

Concentrations of essential trace elements (essential TMEs), including copper (Cu), manganese (Mn), zinc (Zn), and iron (Fe), showed high variability (ranging from high to low or undetectable) throughout the study period in both ponds. Copper was absent in both basins except in the *C. glomerata* pond during autumn, where a concentration of 0.002 ppm was measured. In comparison, manganese levels in the *C. glomerata* pond were approximately double those in *C. vulgaris*, measuring 10.3 ppm in winter and 27.183 ppm in summer, versus 5.06 ppm and 10.34 ppm, respectively, in the *C. vulgaris* pond. As regards zinc, the highest concentrations were recorded in summer, with 0.35 ppm in the *C. glomerata* basin and 0.22 ppm in the *C. vulgaris* basin. The lowest values were observed in winter for the *C. glomerata* basin 0.10 ppm and in spring for the *C. vulgaris* basin

0.02 ppm. The lowest iron concentrations for both basins were observed during the winter season, reaching 0.19 ppm in the *C.vulgaris* pond and 0.11 ppm in the *C.glomerata* pond. Concentrations of potentially toxic trace elements (polluting TMEs), including cadmium (Cd), lead (Pb), nickel (Ni), and aluminium (Al), showed marked temporal and spatial variability throughout the study period in both ponds. The highest concentration of cadmium recorded in the *C.glomerata* pond during the winter was 0.0037 ppm, whereas in the *C.vulgaris* pond, it did not exceed 0.002 ppm in summer. Furthermore, nickel reached a notable concentration of 0.0039 ppm in the *C.vulgaris* pond in spring and 0.014 ppm in the *C.glomerata* pond in summer. As for lead, it recorded a maximum of 0.0032 ppm in the *C.glomerata* basin and 0.009 ppm in the *C.vulgaris* basin, both during the summer period. In addition, both ponds had low iron concentrations throughout the year, with peaks recorded in summer in both ponds. In the *C.glomerata* basin, the iron concentration was 0.33 ppm, while in the other basin it was 0.582 ppm. Aluminium concentrations were significantly higher in the *C.glomerata* pond during the summer 4.21 ppm and winter 1.21 ppm than in the *C.vulgaris* pond, with values of 0.447 ppm and 0.12 ppm, respectively, from summer to winter.

The one-way analysis of variance revealed highly significant seasonal differences for all variables measured in both basins ($p < 0.001$). F-values were found to be generally higher in the *C.glomerata* basin, indicating greater seasonal variability and sensitivity to evaporation and solute concentration processes compared to the *C.vulgaris* basin. As demonstrated in Fig. 3, the results of the principal component analysis (PCA) emphasise the overall structure of the data as well as the primary factors influencing water chemistry in the two basins. Figure 3a presents a comparison between the two basins, while Fig. 3b, c show the results separately. Figure 3b corresponds to the *C.glomerata* basin and Fig. 3c to the *C.vulgaris* basin.

As illustrated in Fig. 3a, PC1 (44.28% – Seasonal gradient) reflects the strength of the annual cycle. The shift of the points from (January/February) to the summer peak (August), indicates that physicochemical variables (temperature, nutrients) evolve synchronously in both ponds. The PC2 axis (34.11% – Pond typology) is the most relevant axis for this study. It shows that, despite the influence of a common climate, the *C.vulgaris* pond and the *C.glomerata* pond retain distinct chemical signatures. Samples from the *C.glomerata* basin (in red) are located in the upper part of the graph, while those from the *C.vulgaris* basin (in blue) are situated in the lower part. This separation indicates that, despite similar climatic conditions, each basin possesses a unique chemical signature. According to Fig. 3b, the analysis of variables, represented by correlation vectors, helps to explain the factors responsible for these variations. Temperature, pH, and conductivity data are orientated towards the upper right quadrant, strongly associating them with the *C.glomerata* basin and the summer months. This suggests that this basin is characterised by warmer and more alkaline conditions, which are favourable to the growth of *C.glomerata*. Conversely, as illustrated in Fig. 3c, the variables associated with mineralisation – notably calcium (Ca), carbonates, and bicarbonates – are concentrated in the lower right quadrant. They are more strongly correlated with the *C.vulgaris* basin, which corresponds to its affinity for hard, mineral-rich waters. Finally, nutrients and certain metals exhibit specific behaviours. Nitrates (NO_3) and phosphorus (P_2O_5) primarily follow the horizontal axis, indicating an increase in their concentration towards summer in both basins. In contrast, elements such as cadmium (Cd), manganese (Mn), and potassium (K) are more strongly associated with the *C.vulgaris* basin, suggesting a greater accumulation or presence in this particular environment.

Discussions

It is acknowledged that certain species of algae inhabiting stagnant water bodies are recognised as specific indicators of ecosystem conditions. Laplace-Treytore et al.²⁷ and Huynh and Serediak⁵¹ have provided practical field guidelines to aid in the accurate identification of these species. Generally, both species are distinguished by their green coloration and its various shades. Nevertheless, the odor they produce serves as a key differentiating feature: *C.glomerata* emits a foul smell, whereas *C.vulgaris* releases an acidic odor reminiscent of sulfur. Additionally, the three in situ parameters – pH, conductivity, and temperature – showed only minor seasonal variations¹⁴. In this study, the monitored parameters demonstrated reduced stability between winter and spring in both types of pond. Both open and covered ponds exhibited similar seasonal trends, with a gradual increase in parameters during the summer, followed by a steady decline from autumn to winter. However, the open pond colonized by *C.vulgaris* generally showed higher values than the covered pond dominated by *C.glomerata*.

A comprehensive analysis of the water quality in the two reservoirs clearly demonstrates that their richness is attributable to water pumped from the region's primary dam, the Youssef Ben Tachafine Dam, and to groundwater, which is already affected by some contamination. It is important to note that the decrease in the volume of water in the dam, when considered in conjunction with the rise in water temperature resulting from global warming and hydrological changes over the last two decades, has led to high mineralisation of the water and enrichment of the water column with nutrients. These nutrients are derived from sedimentation and decomposition. The analysis of the data indicates that the alteration in the trophic status of the dam, which has transitioned from mesotrophic to hypertrophic, can be attributed directly to the observed set of changes³⁴. The second source of water for these reservoirs is groundwater, which is generally of good quality, except in areas where salinity is high or where there are signs of nitrogen pollution from domestic and agricultural sources⁵². In the Oued Issen valley, the coastal area and Chtouka-Aït Baha region, groundwater salinity is moderate. Furthermore, analyses reveal that nitrate levels are moderate to high in Chtouka and in areas contaminated by wastewater and solid waste (*). Consequently, the water renewal rate in the basins depends on the level of these two sources, which has decreased significantly in recent years due to the increasing frequency of droughts. In practice, pumping capacity was only guaranteed at intervals of fifteen days to one month. This irregularity in water renewal directly contributed to the blooms in the basins¹⁴.

Furthermore, the water renewal process is closely linked to the intensity of agricultural activity and the degree of eutrophication of the basin waters. The basins have been developed in accordance with specific standards. The *C.glomerata* basin measures 25 m in length, 10 m in width, and 3 m in depth, while the *C.vulgaris*

basin measures 15 m in length, 7 m in width, and 3 m in depth. The dimensions of the basins are determined by the farm manager, with consideration given to the type of crop, the cost of installation, and the fill rate¹⁴. With regard to the *C. vulgaris* pond, its nutrient richness, higher than that of the second pond, was examined through surveys with the managers of both sites. It was determined that the pond's limited size directly and indirectly contributes to the accumulation of nutrients in its waters. The product's limited dimensions allow for a substantial concentration of these nutrients, while ensuring minimal dissolution. Furthermore, continuous exposure to sunlight leads to regular evaporation of the water volume. It was also noted that soil dissolution contributes to an increase in mineral salt content. This process is exacerbated by the absence of a properly fitted polyethylene geomembrane, which does not adequately cover the edges and bottom of the pond.

Additionally, it should be noted that the quality of the water in the ponds is generally ensured by the introduction of fish-eating algae or by covering the pond⁵³. The effectiveness of this coverage depends on the quality of the nets, and it plays a crucial role in maintaining water quality. The pond, essentially a sheltered wetland where *C. glomerata* naturally grows, is not formally protected. The surrounding net is severely damaged, with tears and holes allowing most of the light to reach the bottom of the pond. This situation is conducive to algal proliferation, which in turn generates a foul odour due to the decomposition of algal biomass. In light of the present circumstances, agricultural producers are unable to replace the damaged nets due to financial constraints. Furthermore, algal proliferation is not perceived as a significant concern, given the ongoing water crisis in the region. The observation was corroborated by the proprietors of the other open pond, who explained their decision not to cover their ponds while increasing the oxygenation rate of their water supply¹⁴. Indeed, the interaction of these parameters within reservoirs has the potential to impact algal blooms from one season to the next.

Indeed, a one-way analysis of variance (ANOVA) was performed in order to assess the effect of season on the physicochemical parameters of the studied basin. The findings indicate a substantial influence of season on the variables examined. The results obtained demonstrate that inter-seasonal variability is significantly greater than intra-seasonal variability, indicating a notable seasonal effect on the physicochemical quality of the water. However, the significant difference between the Fisher test values can be explained by the low dispersion of measurements within each season. This intra-seasonal homogeneity is a key factor that strengthens the F-values. The analysis reveals a clear separation along the vertical axis of Fig. 3a (PC2, explaining 34.11% of the variance), indicating that the two basins exhibit distinct chemical signatures despite similar climatic conditions. The *C. glomerata* basin, positioned in the upper part of the graph, is characterised by higher values of pH, temperature, and conductivity. In contrast, the *C. vulgaris* basin, located in the lower part, is associated with high mineralisation, particularly in carbonates and calcium. Although the similarity in their seasonal cycle, reflected by a left-to-right movement along the PC1 axis, there are significant differences in their chemical responses. The *C. vulgaris* basin appears chemically more stable, as evidenced by the clustering of its points in highly mineralised zones. In contrast, the *C. glomerata* basin shows heightened reactivity, characterised by more pronounced variations, particularly in response to temperature fluctuations and peaks in total dry residue observed during the summer months. An analysis of the correlation arrows indicates that there are different behaviours with respect to metals between the two basins. The *C. vulgaris* basin is distinguished by its notable association with manganese (Mn), cadmium (Cd), and potassium (K), indicating a particular affinity for these elements. Therefore, the *C. glomerata* basin has been found to be more strongly linked to zinc (Zn) and copper (Cu), particularly during the warmer months, reflecting a more sensitive response to seasonal variations in these metals.

Accordingly, the results of the PCA demonstrate that the summer months are characterised by elevated concentrations of nutrients, salts and metals. This phenomenon can be explained by a number of environmental mechanisms, notably the input from already polluted water sources. Firstly, elevated temperatures result in intense evaporation, thereby reducing the water volume and mechanically concentrating dissolved substances, including conductivity, salts (Na, K, Ca, Mg), carbonates and bicarbonates, as well as trace metals. This is a classic effect in semi-arid aquatic environments. Secondly, the reduced dilution caused by summer rainfall contributes to the accumulation of pollutants, in contrast to the natural dilution of dissolved substances that occurs during winter. The warm season also coincides with the intensification of agricultural activities, including the increased use of fertilisers (N, P, K), as well as the return of agricultural runoff to ponds, waters rich in nitrates, phosphates, and trace metals (Cu, Zn). Furthermore, biological mineralisation is enhanced by heat, accelerating the decomposition of organic matter and microbial mineralisation, which in turn releases nutrients, minerals and metals adsorbed in sediments. Consequently, lower water levels and irrigation-induced turbulence promote the mobilisation of sediment, resulting in elevated levels of Fe, Mn, Cu, and Zn in the water column. In summary, the distinction between the two basins is mainly determined by calcium mineralisation, which is more pronounced in the *C. vulgaris* basin, and by thermal and alkaline reactivity, which is stronger in the *C. glomerata* basin, likely due to its coverage. Collectively, these mechanisms elucidate the strong association between PCA and elevated nutrient and metal levels during the summer months, thereby promoting increased pollution and algal blooms in the ponds.

Trace metal elements (TMEs) were analyzed to assess water quality and to elucidate contamination processes and the functioning of the studied aquatic ecosystems. These elements, which occur at very low concentrations, may originate from natural sources such as rock weathering and soils, or from anthropogenic activities including agriculture and domestic or industrial discharges⁵⁴. Trace metals fulfill important ecological functions, some are essential for biological metabolism (Fe, Mn, Zn, Cu, B), while others are toxic even at low concentrations (Cd, Pb, Ni, Al). Accumulation of these metals in aquatic environments can affect macroalgal growth and disrupt the ecological balance of ponds^{55, 56}. Therefore, analyzing TMEs facilitates the identification of potential pollution sources and enhances understanding of the interactions between physicochemical parameters and algal community dynamics in these systems.

This phenomenon is closely linked to the ecophysiological properties of the algal species present. Furthermore, each species has specific requirements regarding light, temperature, nutrient availability, and the physicochemical conditions of the water. Seasonal variations in these factors may create conditions that are either favourable or restrictive for the growth of different species of algae, thereby inducing algal successions throughout the year. These succession patterns are typically characterised by periods of optimal growth, followed by phases of decline, depending on the capacity of species to adapt to environmental fluctuations^{57, 58, 59}. In this regard, *Ulothrix* has a short life cycle, typically ranging from a few days to a few weeks²⁷. In contrast, it has been demonstrated by members of the Characeae family that they possess the capacity for survival and proliferation over extended periods. The maintenance of their biological cycle is influenced by multiple abiotic factors, including the type of aquatic habitat and water quality, which are closely linked to the succession of seasons^{60, 61}. They can have a significant impact on the aquatic ecosystem in which they are found, influencing water quality and transparency⁶². Characeae species, such as *C. vulgaris*, are always affected by chemical pollution and competition from plant species, they are highly sensitive to chemical pollutants in water^{30, 63, 64, 65}, particularly nitrates, phosphates, hydrocarbons, and pesticides, which lead to their disappearance^{32, 66, 67, 68}. However, *Chara* of the genus *vulgaris* can tolerate fairly rich water^{65, 67, 69}, in line with the findings of our study. Characeae can be considered bioindicators sensitive to eutrophication problems^{70, 71}. One study indicated that dissolved organic carbon richness and water transparency influence their growth^{65, 72}, as does calcium concentration⁶⁸. It can thus be posited that the colonisation of *C. vulgaris*, a species which has been demonstrated to withstand temporary desiccation^{73, 74}, may be attributed to its distinct acclimatisation and resilience capacities⁴. This species has been shown to tolerate high levels of UV radiation through the use of protective physiological, morphological, and behavioural processes that optimise photosynthetic activity in low-light conditions^{66, 75}.

Numerous studies have demonstrated a relationship between *Cladophora glomerata* and the level of nutrient input into the environment, thus demonstrating that it only becomes enriched in species when nutrient concentrations increase throughout the gradient^{76, 77}. This genus demonstrates a high level of tolerance to saline environments, withstanding salinity levels of up to 100 ‰, a factor that enhances its ecological value^{23, 78}. Furthermore, the optimal growth of the organism is maintained when the electrolytic conductivity values and the total amount of solutes reach a level of 300–500 µS/cm⁷⁷. While some algal species can develop in nutrient-rich environments⁷⁹, *C. glomerata* has a unimodal distribution⁸⁰, indicating that its presence is closely linked to high water fertility, requiring in particular high concentrations of nitrates and phosphates^{80, 81}, which is consistent with the observations of our study. The combination of different abiotic conditions allows organic spores to be produced from April to the end of August, followed by the development of large quantities of winter resting stages^{82, 83, 84}. In general, *Cladophora* species are among the macroalgae that have become environmental indicators because they are capable of accumulating pollutants (e.g., heavy metals, pesticides, halogenated hydrocarbons)^{85, 86}. They can serve as bioindicators for monitoring water quality and environmental pollution²³.

Conclusions

The eutrophication of the two irrigation basins studied in the Souss-Massa region is attributable to several factors, including increased nutrient availability, which promotes the reproduction and adaptation of organisms to local climatic conditions. The elevated concentrations of nitrates, phosphorus, potassium, magnesium, and manganese are primarily attributable to pre-existing sources of pollution, notably the Youssef Ben Tachafin dam and groundwater. In a semi-arid to arid climate, characterised by strong sunlight, insufficient and non-compliant basin maintenance practices, and sporadic water renewal operations, the proliferation of *Cladophora glomerata* and *Chara vulgaris* has persisted continuously in the studied basins. This study is a continuation of previous research on the physicochemical conditions that promote the proliferation of these two macroalgae. This study therefore constitutes an essential preliminary step for future investigations aimed at identifying, with greater precision, the mechanisms underlying this phenomenon. The year-round physicochemical characterisation of the two basins has made it possible to improve our understanding of eutrophication, which represents a particularly worrying issue given that the water stored in these systems is intended for agricultural irrigation. In order to transform this environmental constraint into a potential agronomic opportunity, it would be highly relevant to progressively extend the scope of investigation to additional irrigation basins. At the same time, urgent action is required to reduce water contamination originating from both dam water and groundwater, especially in a regional context marked by recurrent and severe drought over the last decade. The findings obtained in this work underscore the need to adapt water management strategies and seasonal pollution risk assessment, particularly during the summer period, when the accumulation of nutrients and metals appears to favour both pollution intensification and macroalgal blooms in irrigation waters. Considering the nutrient concentrations recorded in this study, the large-scale cultivation and harvesting of *Cladophora glomerata* and *Chara vulgaris* could be envisaged for a range of future applications. The substantial biomass produced by these species may, in particular, represent a valuable resource for the development of more ecological agricultural practices. In the next stage of the research programme, this biomass will be processed through environmentally friendly extraction methods in order to isolate bioactive compounds with biostimulant and biofertilising properties intended for application to various crops.

Data availability

All data generated or analysed during this study are included in this published article and its supplementary information files. Data are available from the corresponding author, Sara Chachoui.

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Sara CHACHOUI “Corresponding author”: Writing, original draft, methodology, sampling and research, data analysis, revision and editing. Nadiya AMKRAZ: Original draft, supervision, revision and editing. Abdelaziz MIMOUNI: Co-supervision, revision and editing. Hassan BOUBAKER: Conceptualization, original project, revision and editing.

Declarations

Competing interests

The authors declare no competing interests.

Ethical statements

The authors confirm that all research conducted as part of this study adheres to the highest ethical standards. The research presented does not involve human subjects, communities, or data relating to individuals. All methodologies were implemented with respect for the environment and landscapes studied to ensure minimal damage and disruption.

Additional information

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